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FOR ENVIRONMENTAL INTEGRATION**

From Climate Hazards to Financial Stress: Analyzing the Systemic Effects of Extreme Weather Events

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There are several reasons why climate risk can be considered a systemic risk.

Geographical concentration

01

Physical hazards cluster regionally — flood zones, heat-stress belts, coastal exposure. Correlated losses across bank portfolios undermine diversification, the standard defence against idiosyncratic risk.

ECB/ESRB 2021: Climate change may affect around 30% of euro area corporate credit exposures and around 70% of these exposures are concentrated in only 25 banks

Amplification via interconnections

02

Cross-holdings between banks, insurers and investment funds create contagion channels. Fire sales under stress can propagate losses well beyond the initially exposed institutions.

Insurers invest 30% of their money via investment funds — so when funds lose, insurers lose as well

Path dependence

03

Delayed action raises costs non-linearly. Unlike cyclical risks, climate damage is irreversible — each year of inaction locks in higher future losses.

Inaction may cost up to 20% of global GDP by the end of twenty-one hundred

Wrong-way risk in buffers

04

Collateral values and insurance coverage deteriorate precisely when climate shocks are systemic.

Only 35% of climate losses are insured in the EU.

Structural model blindspot

05

No historical precedent means standard VaR models systematically underprice climate risk. Markets cannot efficiently discipline exposure before the shock arrives.

Climate risk emerges mainly in the future. Standard models using past data offer limited use for this risk type.

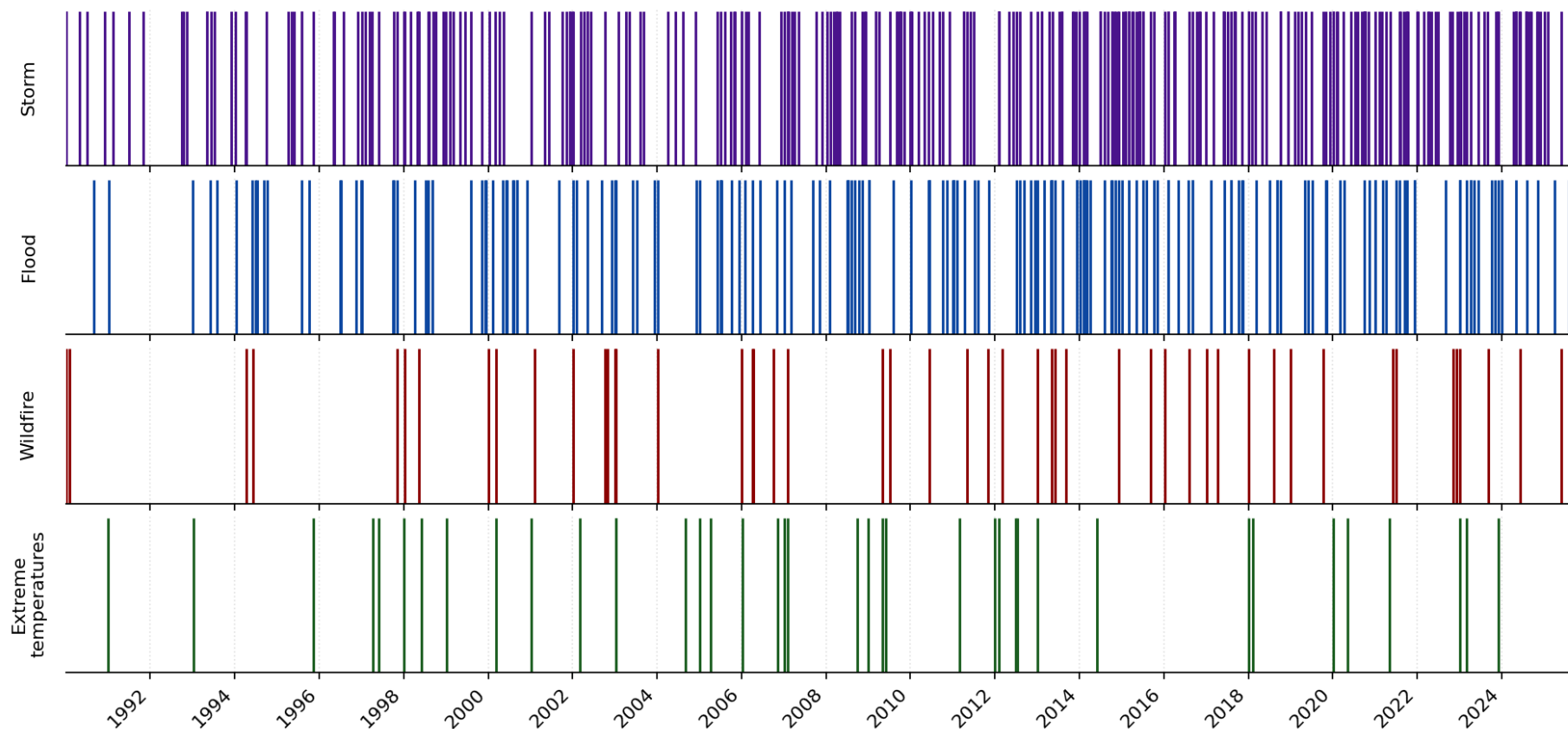
Climate risk can manifest as systemic risk

Aim of the research

The objective of this study is to examine whether **extreme weather events** - specifically, storms, floods, wildfires and extreme temperature episodes - are associated with systemic financial stress.

- Dataset: 1,466 extreme weather events across 11 countries, with daily observations from January 1990 to August 2025.
- When selecting extreme weather events, we used the Emergency Events Database (EM-DAT), a joint initiative of the Centre for Research on the Epidemiology of Disasters and the World Health Organization (Delforge et al., 2025).
- As a baseline for our analysis, we focus on stronger extreme weather events, defined as those causing at least 30 fatalities or generating CPI-adjusted economic costs of at least USD 1 million.

Extreme weather events



809 storms, 425 floods, 105 wildfires and 127 extreme temperature events

CISS as the primary measure

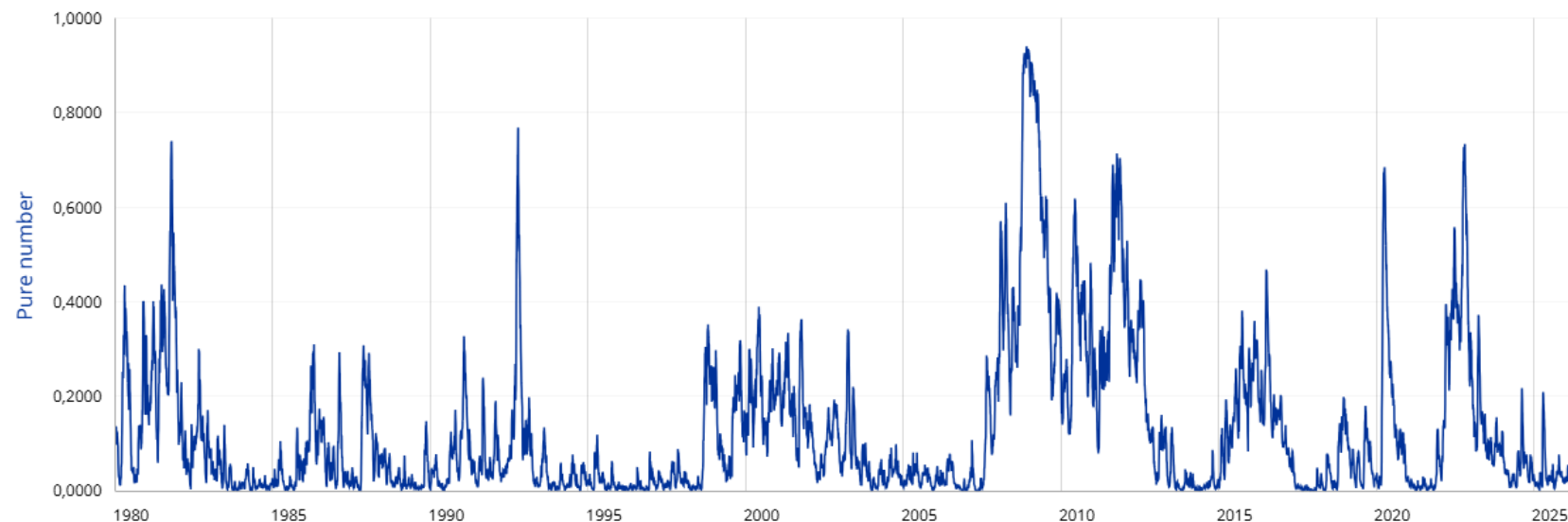
The objective of this study is to examine whether extreme weather events - specifically, storms, floods, wildfires and extreme temperature episodes - are associated with **systemic financial stress**.

- CISS (Composite Indicator of Systemic Stress) as the primary measure of systemic financial stress (Holló et al., 2012).
- The CISS is based on 15 raw financial stress indicators that are predominantly market-based. These indicators are evenly allocated across five categories: the financial intermediary sector, money markets, equity markets, bond markets, and foreign exchange markets.
- The daily CISS levels for selected countries were obtained from the ECB Statistical Data Warehouse.

CISS as the primary measure

ECB Data Portal, 1 July 2026, 10:24 CET

■ NEW CISS - Composite Indicator of Systemic Stress., Euro area, Daily



Source: ECB

EUROPEAN CENTRAL BANK | EUROSISTEM

<https://data.ecb.europa.eu>

- CISS ranges from 0 to 1.
- Higher values indicate higher systemic financial stress.
- Large increases are visible during the Global Financial Crisis, the euro area debt crisis, COVID-19 and major geopolitical shocks.

Methodology

For each event category and time window, we estimate the following:

$$\Delta Y_{i,t} = \alpha + \beta_k D_{i,t}^{(k)} + \gamma Z_{i,t} + u_i + \lambda_t + \epsilon_{i,t}$$

where:

$\Delta Y_{i,t}$ – daily change in the Composite Indicator of Systemic Stress for country i and day t ; for robustness checks, we also use the FSI index (Financial Stress Index).

$D_{i,t}^{(k)}$ - event indicator equal to one for the event day (t_0) or the cumulative postevent window (L1, L2, L3).

$Z_{i,t}$ - matrix of control variables (macroeconomic data releases and interest rate changes).

u_i - country-specific fixed effects.

λ_t - time fixed effects.

$\epsilon_{i,t}$ – error term.

Methodology

For each event category and time window, we estimate the following:

$$\Delta Y_{i,t} = \alpha + \beta_k \textcolor{red}{Ds}_{i,t}^{(k)} + \gamma Z_{i,t} + u_i + \lambda_t + \epsilon_{i,t}$$

where:

$\textcolor{red}{Ds}_{i,t}^{(k)}$ - event severity indicator

$$F_n(x_C) := \begin{cases} \frac{r}{n} & \text{for } x_{[r]} \leq x_C \leq x_{[r+1]}, \quad r = 1, 2, \dots, n-1 \\ 1 & \text{for } x_C = x_{[n]} \end{cases}$$

In addition to the occurrence of extreme weather events, their **severity** is also important. In this study, the severity is measured either by the number of fatalities resulting from the extreme event or by the amount of the CPI-adjusted economic costs associated with the event. Both variables are normalized via the cumulative empirical distribution function (CDF). The severity of the event is defined as the higher CDF value between these two measures.

Results

Weather events	Storm	Storm	Storm	Storm	Flood	Flood	Flood	Flood
Window	t0	L1	L2	L3	t0	L1	L2	L3
macro controls	0.0452** (0.0173)	0.0452** (0.0173)	0.0451** (0.0172)	0.0452** (0.0173)	0.0451** (0.0173)	0.0451** (0.0172)	0.0453** (0.0172)	0.0452** (0.0171)
IR_change	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1547*** (0.0331)	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1549*** (0.0330)	0.1549*** (0.0330)	0.1548*** (0.0331)
Storm	0.0227 (0.0492)	0.0153 (0.0271)	0.0165 (0.0299)	0.0193 (0.0242)				
Flood					-0.0467 (0.0813)	-0.0709 (0.0938)	-0.0987 (0.1115)	-0.0503 (0.0881)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	88176	88186	88186	88186	88186	88186	88186	88186

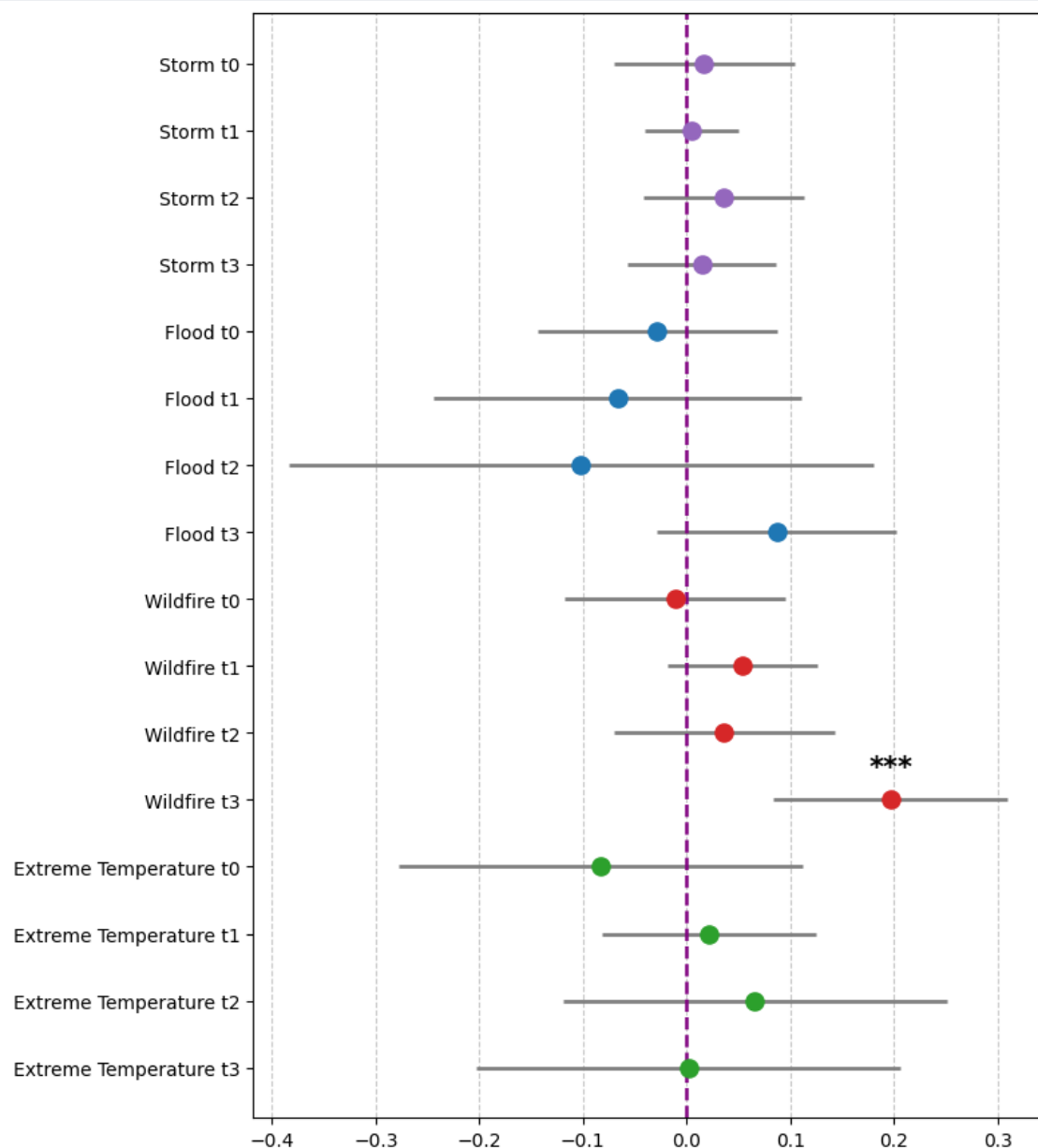
Weather events	Wildfire	Wildfire	Wildfire	Wildfire	Extreme temp	Extreme temp	Extreme temp	Extreme temp
Window	t0	L1	L2	L3	t0	L1	L2	L3
macro controls	0.0451** (0.0172)	0.0453** (0.0173)	0.0452** (0.0173)	0.0454** (0.0173)	0.0452** (0.0173)	0.0452** (0.0173)	0.0451** (0.0173)	0.0451** (0.0173)
IR_change	0.1547*** (0.0331)	0.1548*** (0.0332)	0.1548*** (0.0331)	0.1551*** (0.0332)	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1547*** (0.0331)	0.1547*** (0.0331)
Wildfire	-0.0001 (0.0629)	0.0291 (0.0353)	0.0282 (0.0279)	0.0780* (0.0311)				
Extreme temp					-0.0778 (0.1038)	-0.0291 (0.0700)	0.0097 (0.0644)	-0.0064 (0.0624)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	88176	88176	88176	88176	88176	88176	88176	88176

Results

Weather events	Storm_s	Storm_s	Storm_s	Storm_s	Flood_s	Flood_s	Flood_s	Flood_s
Window	t0	L1	L2	L3	t0	L1	L2	L3
macro controls	0.0452** (0.0173)	0.0452** (0.0172)	0.0451** (0.0172)	0.0452** (0.0173)	0.0451** (0.0173)	0.0451** (0.0172)	0.0453** (0.0172)	0.0453** (0.0171)
IR_change	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1547*** (0.0331)	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1549*** (0.0330)	0.1549*** (0.0330)	0.1547*** (0.0331)
Storm	0.0180 (0.0590)	0.0170 (0.0316)	0.0163 (0.0342)	0.0158 (0.0280)				
Flood					-0.0509 (0.0887)	-0.0795 (0.1021)	-0.1093 (0.1166)	-0.0541 (0.0904)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	88176	88186	88186	88186	88186	88186	88186	88186

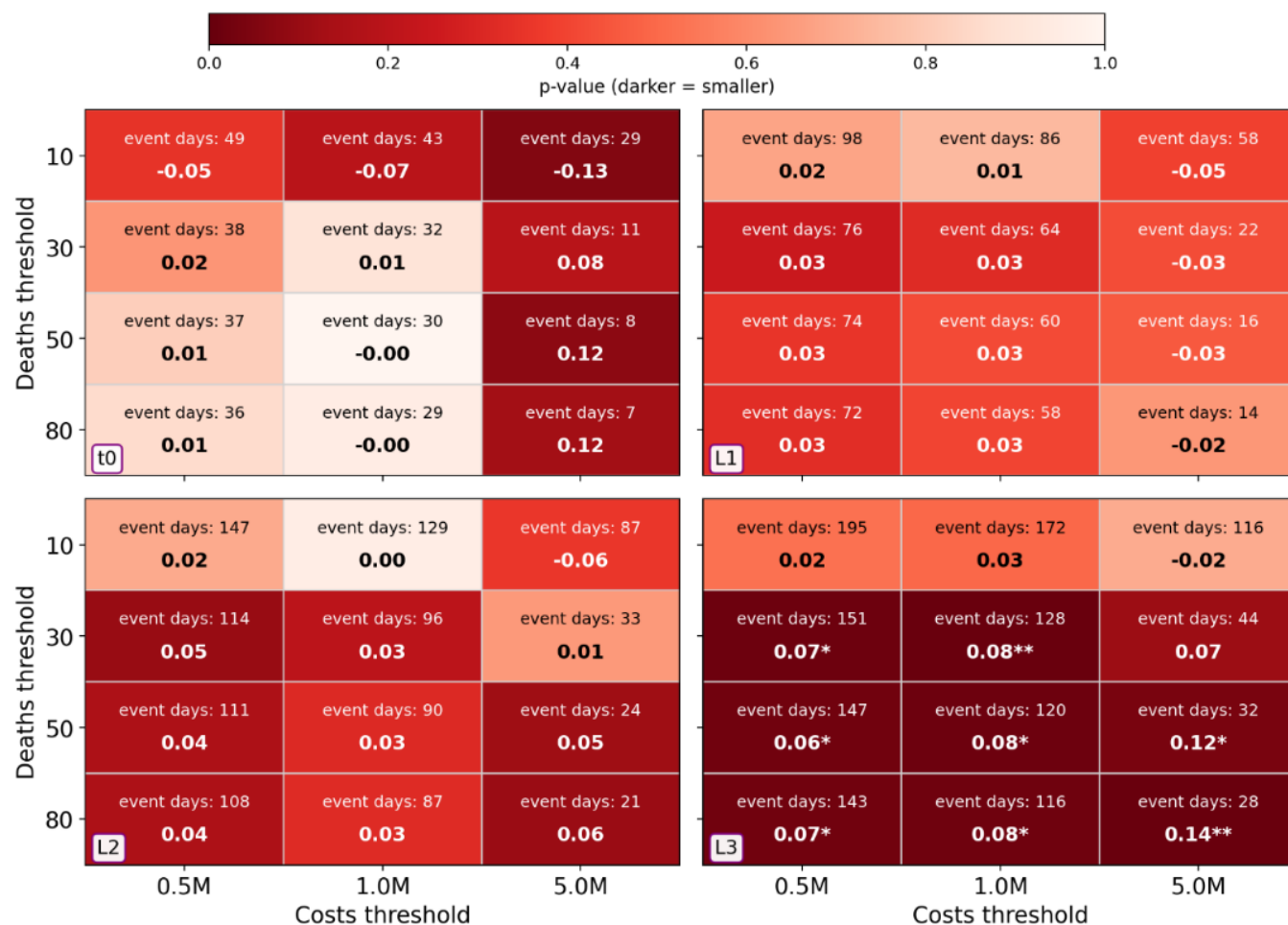
Weather events	Wildfire_s	Wildfire_s	Wildfire_s	Wildfire_s	Extreme temp_s	Extreme temp_s	Extreme temp_s	Extreme temp_s
Window	t0	L1	L2	L3	t0	L1	L2	L3
macro controls	0.0451** (0.0172)	0.0453** (0.0173)	0.0452** (0.0173)	0.0454** (0.0173)	0.0452** (0.0173)	0.0452** (0.0173)	0.0451** (0.0173)	0.0451** (0.0173)
IR_change	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1548*** (0.0331)	0.1551*** (0.0332)	0.1547*** (0.0331)	0.1548*** (0.0331)	0.1547*** (0.0331)	0.1547*** (0.0331)
Wildfire	-0.0039 (0.0686)	0.0282 (0.0389)	0.0285 (0.0314)	0.0858* (0.0345)				
Extreme temp					-0.0849 (0.1315)	-0.0282 (0.0876)	0.0143 (0.0804)	-0.0102 (0.0778)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	88176	88176	88176	88176	88176	88176	88176	88176

Results – daily effect



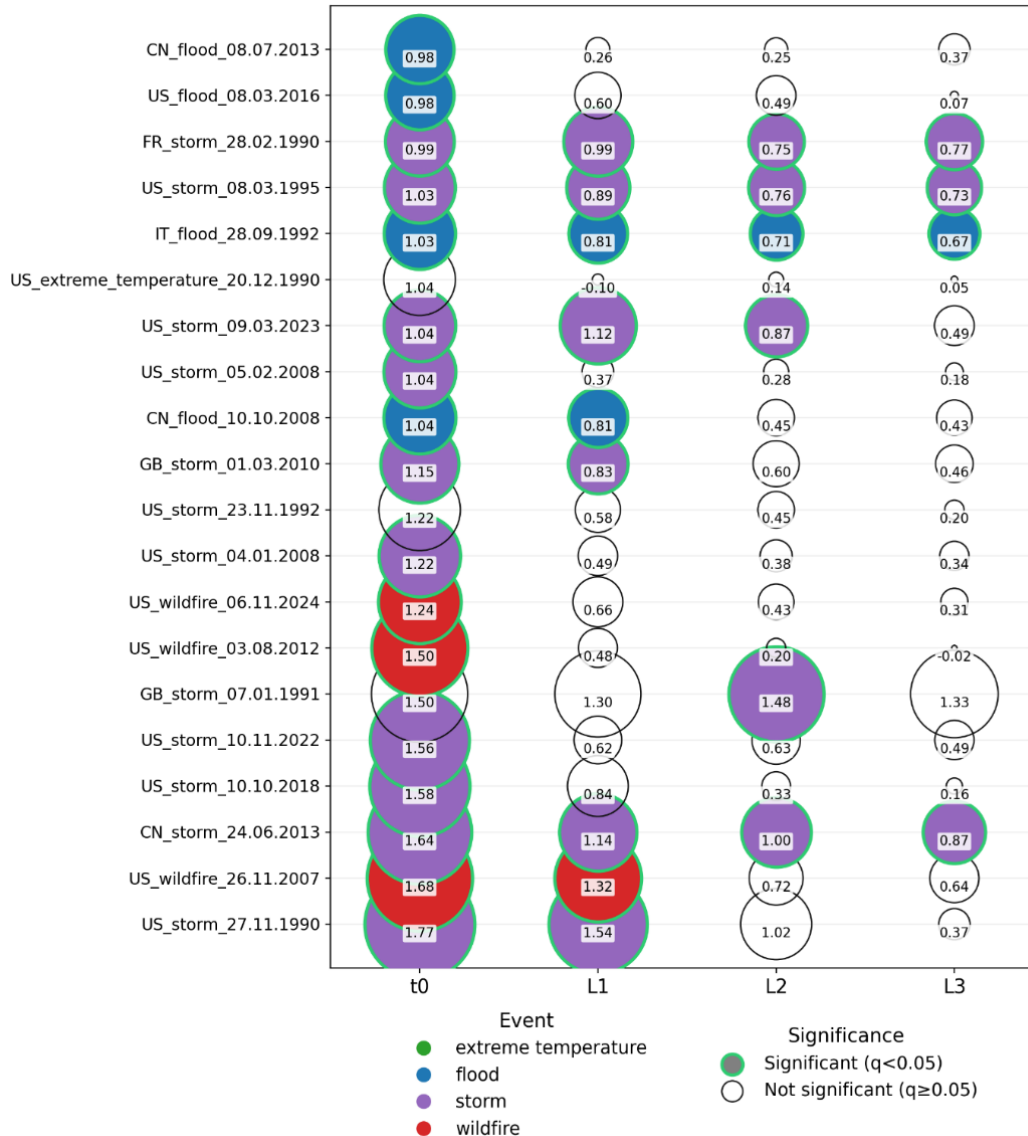
- Previous results used **cumulative event windows**.
- This figure uses **day-by-day windows** to measure the effect on each day.
- The results confirm the previous analysis. Even with day-by-day windows, only wildfires have a significant effect. The effect appears on the third day after the event. Storms, floods, and extreme temperature events are still not significant.

Robustness check



- In our baseline specification, we selected events causing at least 30 fatalities or generating CPI-adjusted economic losses of at least USD 1 million. However, we also performed robustness checks using different event selection criteria.
- The results remain consistent across different specifications.
- Only wildfires show a significant effect in the L3 window.

Most severe events



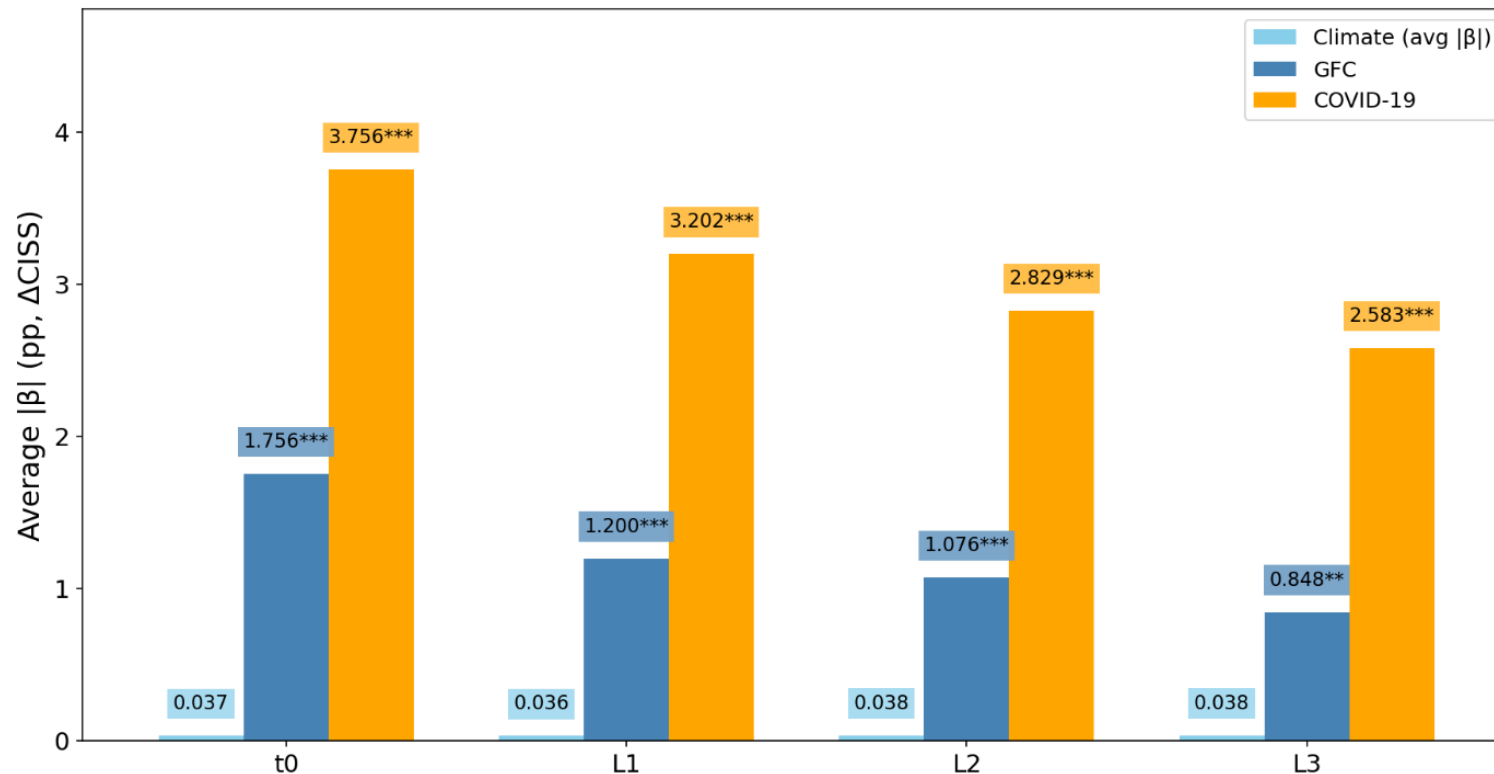
- Among the 20 most severe events, storms and wildfires clearly dominate.
- The most severe cases are the U.S. storm of November 1990 and the U.S. wildfire of 2007, both of which generated relatively large systemic stress responses.
- The coefficient of 1.77 for the 1990 U.S. storm indicates that after global and country-level dynamics are filtered out, the event-day residual change in systemic stress was almost 1.8 units greater than its expected daily change.
- 27 November 1990 O'Fallon, USA (Illinois) severe tornado

Anticipation effects



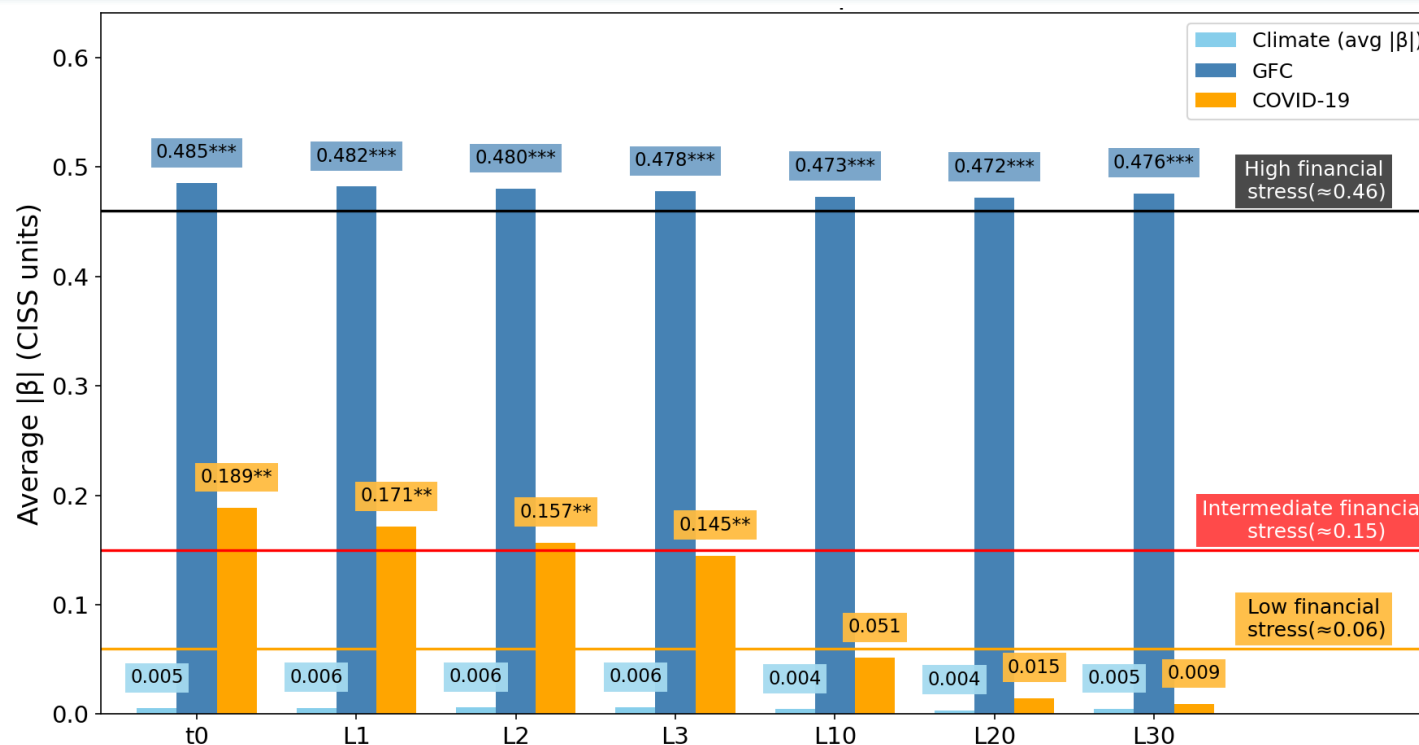
- Investor reactions often start in response to weather forecasts alone. This was clearly demonstrated during Hurricane Katrina, when investors reacted before the hurricane arrived (Williams and Bacon, 2021).
- This motivates us to test for potential anticipation effects.
- The results suggest that anticipation effects are limited. We can see them only in a few individual cases. For most storms, there is no significant market reaction before the event.

Change in Systemic Financial Stress: Climate Events vs. Major Financial Crises



- This figure compares the average impact of climate events with two major financial crises.
- Climate events have a much smaller impact than major financial crises.
- GFC has a much stronger impact. COVID-19 has the largest impact.

Level of Systemic Financial Stress: Climate Events vs. Major Financial Crises



Note: Following Holló et al. (2012), systemic stress can be classified into three regimes: low stress ≈ 0.06 , intermediate stress ≈ 0.15 , crisis (high) stress > 0.46 .

- This figure compares the **level** of financial stress after climate events, the Global Financial Crisis, and COVID-19
- COVID-19 reached moderate financial stress.
- The Global Financial Crisis reached high financial stress.

Conclusions

- Wildfires have the strongest impact on systemic financial stress. The effect appears about three days after the event. The results are robust.
- Other types of extreme weather events (storms, floods, and extreme temperature events) have not shown a significant impact on systemic financial stress (so far).
- Climate events generate much less systemic financial stress than major financial crises.
- It is important to remember that we study systemic financial stress. Our results do not mean that extreme weather events have no impact on the financial system. Previous studies, for example Kurowski and Rogowicz (2026), show that some parts of the financial system, such as financial markets, can be affected much more strongly.



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Thank you!

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